

Bayes Filter – Discrete Filters

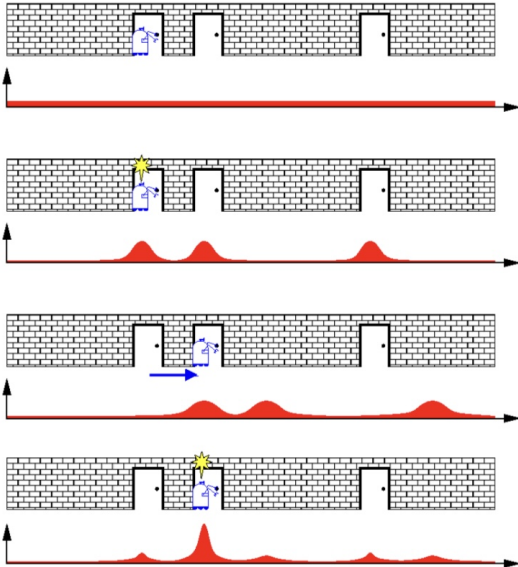
Autonomous Navigation of Mobile Robots

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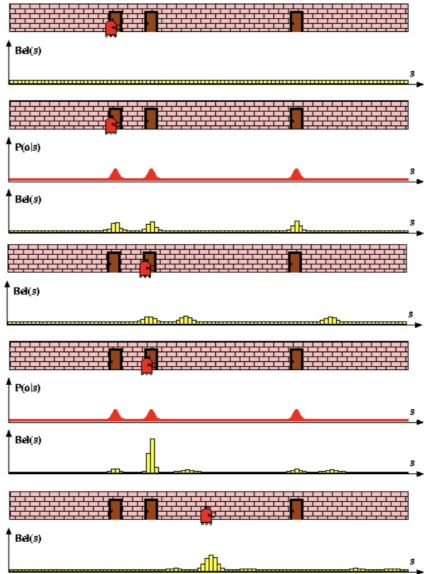
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Probabilistic Localization



Piecewise Constant



Discrete Bayes Filter Algorithm

Algorithm `discrete_bayes_filter` ($Bel(x), d$) :

$\eta = 0$

If d is a **perceptual** data item z Then

for all x do

$$Bel'(x) = P(z | x) Bel(x)$$

$$\eta = \eta + Bel'(x)$$

for all x do

$$Bel'(x) = \eta^{-1} Bel'(x)$$

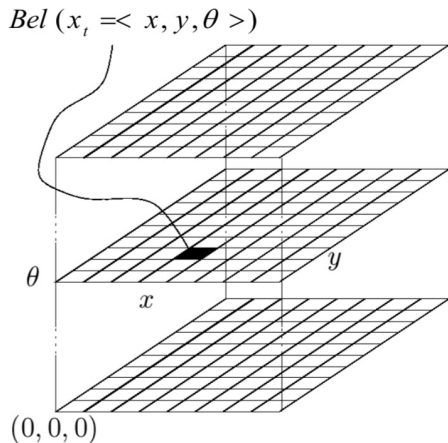
Else if d is an **action** data item u Then

for all x do

$$Bel'(x) = \sum_{x'} P(x | u, x') Bel(x')$$

Return $Bel'(x)$

Piecewise Constant Representation



Implementation (1)

- To update the belief upon sensory input and to carry out the normalization one has to iterate over all cells of the grid
- Especially when the belief is peaked (which is generally the case during position tracking), one wants to avoid updating irrelevant aspects of the state space
- One approach is not to update entire sub-spaces of the state space
- This, however, requires to monitor whether the robot is de-localized or not
- To achieve this, one can consider the likelihood of the observations given the active components of the state space

Implementation (2)

- To efficiently update the belief upon robot motions, one typically assumes a bounded Gaussian model for the motion uncertainty
- This reduces the update cost from $O(n^2)$ to $O(n)$, where n is the number of states
- The update can also be realized by shifting the data in the grid according to the measured motion
- In a second step, the grid is then convolved using a separable Gaussian Kernel
- Two-dimensional example

1/16	1/8	1/16
1/8	1/4	1/8
1/16	1/8	1/16

 $\cong$

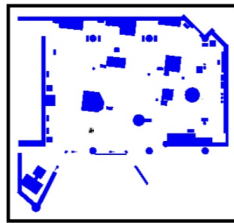
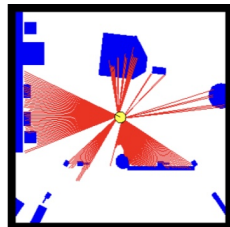
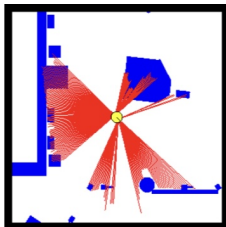
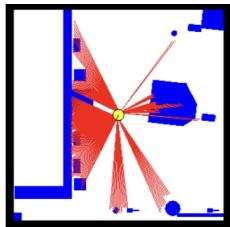
1/4
1/2
1/4

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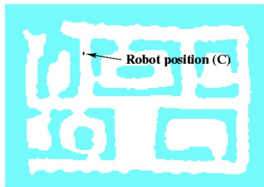
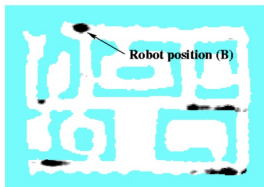
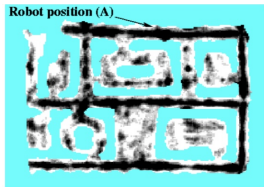
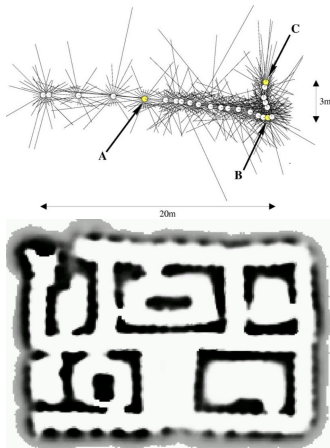
1/4	1/2	1/4
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- Fewer arithmetic operations, easier to implement

Grid-based Localization

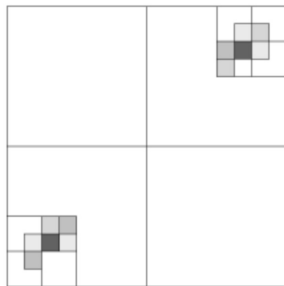
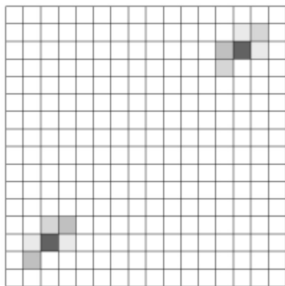


Sonars and Occupancy Grid Map



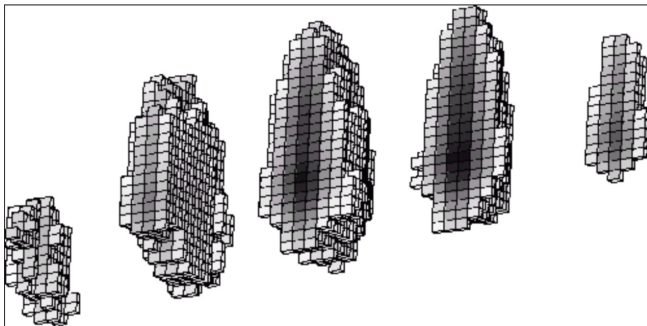
Tree-based Representation

Idea: Represent density using a variant of octrees



Tree-based Representation

- Efficient in space and time
- Multi-resolution



Summary

- Discrete filters are an alternative way for implementing Bayes Filters
- They are based on histograms for representing the density
- They have huge memory and processing requirements
- Can easily recover from localization errors
- Their accuracy depends on the resolution of the grid
- Special approximations need to be made to make this approach having dynamic memory and computational requirements